Module 8: NLP Preprocessing I

Learning Objectives

- Understand NLP preprocessing concepts
- Apply tokenization, stop-word removal
- Compare stemming vs lemmatization
- Build preprocessing pipelines

1. Introduction (Aligned with Indurkhiya & Damerau Ch.2)

Text preprocessing (normalization) prepares raw text for NLP tasks.
Key ideas:
- Normalize text (case folding, punctuation removal)
- Reduce linguistic variability
- Prepare for feature extraction

2. What is NLP Preprocessing?

Transform raw text into clean, structured data.

Example Problem:

"I LOVE NLP!!! It's amazing!!!"

Issues:

- Uppercase
- Punctuation
- Noise

3. Tokenization

Tokenization splits text into smaller units.

Diagram:
Raw Text → [Tokenizer] → Tokens

Example:

```
from nltk.tokenize import word_tokenize
```

```
text = "NLP is powerful."
tokens = word_tokenize(text)
print(tokens)
```

Output:

['NLP', 'is', 'powerful', '.']

4. Stop-word Removal

Removes high-frequency words that carry little meaning.

Diagram:
Tokens → [Stopword Filter] → Meaningful Tokens

Note (Textbook Insight):
Some stopwords may carry meaning in context (e.g., 'not').

```
from nltk.corpus import stopwords

stop_words = set(stopwords.words('english'))
tokens = ['nlp', 'is', 'fun']
filtered = [w for w in tokens if w.lower() not in stop_words]
print(filtered)
```

Output:

['nlp', 'fun']

5. Stemming

Stemming reduces words to crude root forms.

Diagram:
running → run
studies → studi

Fast but less accurate.

```
from nltk.stem import PorterStemmer

stemmer = PorterStemmer()
```

```
words = ["running", "studies"]
print([stemmer.stem(w) for w in words])
```

Output:

run study

6. Lemmatization

Lemmatization uses vocabulary + morphology. It converts words to dictionary form.

Diagram:
running → run
better → good

More accurate but slower.

```
from nltk.stem import WordNetLemmatizer

lemmatizer = WordNetLemmatizer()
words = ["running", "better"]
print([lemmatizer.lemmatize(w, pos='v') for w in words])
```

```
from nltk.stem import WordNetLemmatizer
lemmatizer = WordNetLemmatizer()
print(lemmatizer.lemmatize("better", pos='v'))
```

Output:

good

7. Stemming vs Lemmatization

Stemming:
- Fast
- Less accurate

Lemmatization:
- Slower
- Linguistically correct

8. NLP Pipeline (End-to-End)

Diagram:
Raw Text
↓
Cleaning
↓
Tokenization
↓
Stopword Removal
↓
Stemming/Lemmatization
↓
Processed Text

```
import re
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords

text = "NLP is amazing! It helps machines understand humans."

text = re.sub(r'[^a-zA-Z\s]', '', text.lower())
tokens = word_tokenize(text)

stop_words = set(stopwords.words('english'))
tokens = [w for w in tokens if w not in stop_words]

print(tokens)
```

Notes

- Ensure NLTK resources are downloaded
- Emphasize context in stopword removal
- Compare outputs of stemming vs lemmatization